Khushaan Virk

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Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science (BASc) in Electrical Engineering — *Expected May 2028*

- **Relevant Coursework:** Circuit Analysis, Digital Logic Design, Signals & Systems, Embedded Systems Design

Experience

Software Engineer — Jan. 2026 – Present
Advizr — *Vancouver, BC*

- Built Python backend services integrating Claude and OpenAI APIs to automate lead generation, content workflows, and operational pipelines for 3 clients across real estate, fintech, and healthcare
- Shipped FastAPI backends and scheduled automation pipelines on Railway with Supabase as the data layer and Vercel-hosted client frontends, eliminating 30+ hours/week of manual workflows across deployed client systems
- Owned client engagements end to end from open-ended business goals to production, including a Canada-wide lead generation engine producing 11,000+ leads with on-demand rescraping

Sports Instructor / Summer Camp Leader — Sep. 2023 – Sep. 2025
City of Surrey — *Surrey, BC*

- Led structured programs for groups of 15–30, coordinating daily operations across staff and volunteers while maintaining safety standards
- Adapted instruction in real time to participants of varying skill levels, communicating clear expectations to children, parents, and co-instructors while resolving conflicts and incidents on the spot

Projects

Pedestrian Dead Reckoning (Phone IMU) | *Python, NumPy, SciPy, pandas, Matplotlib* — Jul. 2026

- Built a Python signal-processing pipeline in NumPy and SciPy to reconstruct a walking path from phone accelerometer and gyroscope logs alone, band-passing acceleration magnitude to the 0.7–2.1 Hz gait band and running peak detection for footfall counting, with cadence cross-checked against an FFT of the walk
- Estimated gyroscope bias from a stationary window, integrated the corrected rate for heading, and dead-reckoned an (x, y) trajectory step by step, measuring inertial drift by loop closure over a 76.2 m route at 5.1 to 6.0 m, roughly 7% of distance walked, with step length held out across two runs to avoid overfitting

Field-Following Autonomous Robot | *STM32, C, MOSFETs, I2C, IR* — Apr. 2026

- Built a dual-microcontroller robot with MOSFET H-bridge motor drivers and opto-isolated control lines for electromagnetic noise reduction during inductive sensor readings
- Implemented magnetic field detection using pick-up inductors to autonomously follow a guide wire, with a third sensor for intersection detection and pre-configured path navigation
- Designed an IR remote controller on a separate MCU transmitting 38kHz modulated commands for manual control and path selection
- Integrated VL53L0X Time-of-Flight laser ranging modules over I2C for real-time collision detection and avoidance in autonomous mode

Reflow Oven Controller | *Op-Amp, ADC, SSR, FSM, Python* — Feb. 2026

- Built a Python serial data acquisition framework around the firmware: real-time strip-chart plotting and automated Excel report generation over UART at 115200 baud, used as the verification harness during firmware tuning
- Wrote interrupt-driven embedded firmware implementing a 5-state FSM with PWM-based SSR control for precise thermal profile tracking through preheat, soak, reflow, and cooling phases
- Designed a closed-loop temperature control system using a K-type thermocouple with cold-junction compensation (LM335) and OP07 op-amp signal conditioning, achieving ±3°C accuracy over 25–240°C
- Integrated LCD/7-segment displays, speaker feedback, and safety logic with automatic abort on sensor failure

FPGA Tron Game | *DE10-Lite, Nios V (RISC-V), C, VGA* — Nov. 2025

- Built a real-time two-player Tron lightcycle game on a DE10-Lite FPGA with game logic in C on a Nios V (RISC-V) soft core, demonstrating end-to-end RTL-to-software integration
- Implemented a custom double-buffered VGA renderer (bit masking on memory-mapped control registers, off-screen compose, screen-refresh swap) with hardware-interrupt-driven inputs to hold tear-free 60 Hz output

Technical Skills

Embedded & Control Systems: ARM (STM32), FPGA (DE10-Lite, Nios V soft core), ADC, I2C, SPI, UART, PWM
Programming: Python, C, Embedded C, SystemVerilog, SQL
Instrumentation & Hardware: Oscilloscopes, multimeters, logic analyzers, soldering, signal conditioning, PCB layout
Tools & Software: Git, KiCad, EasyEDA, MATLAB, serial communication (PuTTY), CrossIDE